

NODE 4 — INTRALUMINAL GEOMETRY, COMPLIANCE, AND CONTACT DEVICE WITH CONTROLLED RADIAL STIMULUS

A. Overview

In certain embodiments, a fourth node is configured as an intraluminal device configured to measure internal geometry, compliance, and contact characteristics using endpoint sensing and controlled radial stimulus. The fourth node may be configured for intravaginal use and/or anal use, thereby supporting a range of users and pairing contexts. The fourth node may be used in a guided session to generate repeatable, privacy-preserving internal metrics suitable for baseline establishment, longitudinal comparison, and cross-user correlation.

B. Form Factor and Intraluminal Placement

The fourth node may be implemented as an insertable body having a soft external surface and an internal structural framework. The device may include an insertion geometry configured for comfortable placement and stable positioning, including one or more of tapered ends, segmented compliance regions, and axial stabilization features. In certain embodiments, the device includes a handle element, grip region, base flange, or docking base to support controlled insertion depth, repeatable orientation, and user guidance. The housing may be sealed for fluid resistance and cleaning, and may include a retrieval feature.

C. Endpoint and Geometry Sensing

The fourth node may include one or more endpoint sensors configured to determine internal distance and geometry parameters. Endpoint sensing may be implemented using infrared time-of-flight sensors, optical proximity sensors, and/or other distance sensing modalities. Endpoint sensors may be oriented axially and/or radially, and may be arranged to measure depth, internal diameter, and/or cross-sectional profile at one or more positions. In some embodiments, multiple endpoint sensors provide redundancy and enable shape profiling across angles.

D. Controlled Radial Stimulus Mechanisms

In certain embodiments, the fourth node includes a controlled radial stimulus mechanism configured to apply controlled radial pressure and/or controlled radial expansion to tissue. Example mechanisms include: 1. an annular inflatable cuff or microbladder ring, 2. a constraint band configured to limit expansion, 3. a manual pneumatic actuation mechanism including a compressible handle element, bulb, or reservoir, 4. a micro-pump and valve system, and/or 5. a mechanical cam or other mechanical expansion actuator. Stimulus magnitude may be controlled in open-loop or closed-loop manner. In certain embodiments, stimulus is applied in discrete steps, ramps, or cycles, and the system records response curves and stability measures.

E. Stimulus Pressure Sensing and Contact Confirmation

The fourth node may include one or more stimulus pressure sensors including MEMS pressure transducers to measure applied pressure and to support closed-loop control. The fourth node may include one or more contact sensing elements configured to measure circumferential contact and compression patterns, including one or more of circumferential pressure rings, piezoresistive or FSR rings, capacitive pressure rings, piezoresistive film bands, and/or capacitive proximity/contact confirmation electrodes. In certain

embodiments, capacitive contact confirmation is used to verify valid tissue contact and to support wear validation and anti-spoof checks.

F. Motion Sensing, Robustness, and Artifact Rejection

The fourth node may include motion and orientation sensing comprising a 6-axis IMU to measure motion state and orientation and to support robustness, artifact rejection, and spoof resistance. The computing system may compute a quality index based on motion state, contact stability, sensor agreement, and stimulus consistency. Invalid segments may be excluded from analysis.

G. Derived Features and Metrics

The computing system may derive internal geometry and compliance features including endpoint-derived distance measures, cross-sectional estimates, compliance slopes, threshold measures, hysteresis measures, stability measures, and contact distribution measures. Metrics may be presented as privacy-preserving outputs including indices, zones, ranges, and curves. Example outputs include: an internal geometry profile derived from endpoint sensing, a compliance index derived from response to controlled radial stimulus, a contact stability score derived from circumferential sensing during stimulus, and a confidence indicator derived from motion gating and sensor agreement.

H. Cross-Device and Cross-User Correlation

Node 4 outputs may be correlated with outputs from other nodes, including external geometry metrics, longitudinal physiology metrics, and upper-body context metrics. In certain embodiments, Node 4 outputs are used for cross-user correlation across a range of pairing contexts including male-to-female, male-to-male, female-to-female, and other pairings, using privacy-preserving metrics and without explicit anatomical imagery or explicit anatomical labeling.

I. Variations and Alternatives

The fourth node may be implemented with different stimulus mechanisms, different endpoint sensor types, and different ring or band electrode patterns. Certain embodiments omit active stimulus and use passive contact mapping; other embodiments emphasize multi-step stimulus profiling. Power may be provided via rechargeable battery, inductive charging, or other subsystems. Processing may be performed locally or via a computing system. The device may include a docking base for charging, cleaning, storage, or calibration.